TASK:1

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

The given network address is **192.168.10.0/24**. It can be divided into four equal subnets by borrowing 2 bits from the host portion.

Subnet mask: 255.255.255.192
CIDR notation: /26
Total IP addresses per subnet: 64
Usable host IP addresses per subnet: 62

Subnet	Network Address	Subnet Mask	Usable IP Address Range	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.193 – 192.168.10.254	192.168.10.255

Result: The original /24 network is divided into four equal subnets, each containing 64 total addresses and 62 usable host addresses.

NETWORK

TASK: 2

**For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.**

Assign valid IP addresses to five devices.

For this example, I have selected **Subnet 1: 192.168.10.0/26**.

Subnet mask: 255.255.255.192
Network address: 192.168.10.0
Broadcast address: 192.168.10.63

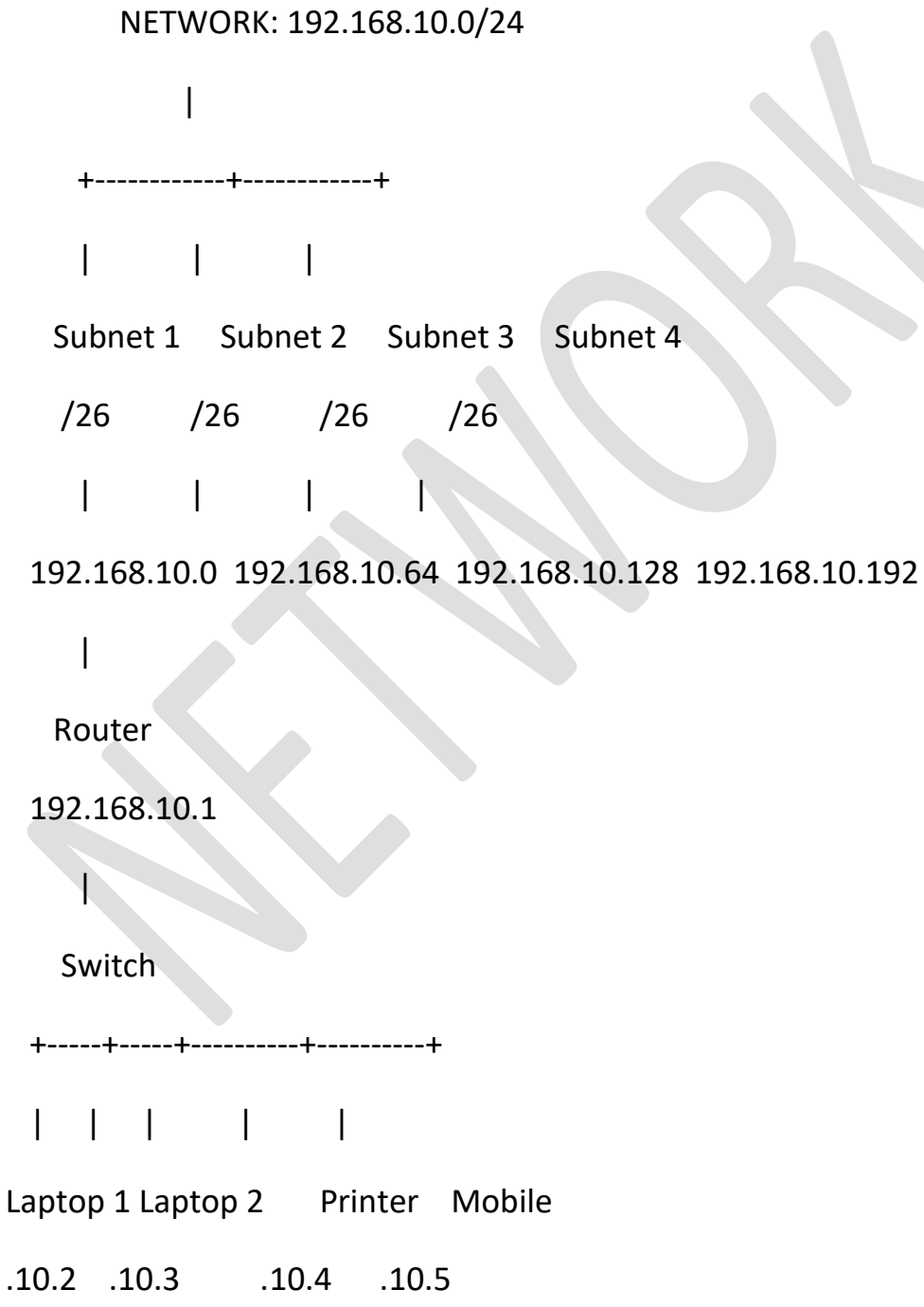
Device	Assigned IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile Phone	192.168.10.5	255.255.255.192

Result: All five devices have unique and valid IP addresses within Subnet 1. The network address (192.168.10.0) and broadcast address (192.168.10.63) are not assigned to any device.

NETWORK

TASK: 3

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



NETWORK

SUBNET DETAILS

Subnet 1: 192.168.10.0/26

Usable IPs: 192.168.10.1 - 192.168.10.62

Broadcast: 192.168.10.63

Subnet 2: 192.168.10.64/26

Usable IPs: 192.168.10.65 - 192.168.10.126

Broadcast: 192.168.10.127

Subnet 3: 192.168.10.128/26

Usable IPs: 192.168.10.129 - 192.168.10.190

Broadcast: 192.168.10.191

Subnet 4: 192.168.10.192/26

Usable IPs: 192.168.10.193 - 192.168.10.254

Broadcast: 192.168.10.255

NETWORK

TASK: 4

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1: Identify the original network.

The given network address is 192.168.10.0/24. The /24 prefix means that 24 bits are used for the network portion and 8 bits are available for host addresses.

Step 2: Calculate the number of bits to borrow.

We need four equal subnets. The formula is:

Number of subnets = 2^n

Here, *n* is the number of bits borrowed from the host portion.

$2^2 = 4$ subnets.

Therefore, we borrow 2 bits.

Step 3: Calculate the new subnet mask.

The original prefix is /24. After borrowing 2 bits:

New prefix = 24 + 2 = **/26**

The subnet mask for /26 is **255.255.255.192**.

Step 4: Calculate the number of addresses per subnet.

There are 32 bits in an IPv4 address. With a /26 prefix, 6 bits remain for hosts.

Number of addresses = $2^6 = 64$.

Usable host addresses = 64 − 2 = **62**.

Two addresses are reserved: one for the network address and one for the broadcast address.

NETWORK

Step 5: Calculate the subnet ranges.

The block size is calculated as:

256 − 192 = **64**

Therefore, the subnet network addresses increase by 64 in the last octet:

- Subnet 1: 192.168.10.0
- Subnet 2: 192.168.10.64
- Subnet 3: 192.168.10.128
- Subnet 4: 192.168.10.192

The broadcast address of each subnet is one less than the next subnet's network address. The last subnet ends at 192.168.10.255.

Step 6: Assign IP addresses to devices.

I selected Subnet 1, which has usable IP addresses from 192.168.10.1 to 192.168.10.62.

I assigned 192.168.10.1 to the router, 192.168.10.2 to Laptop 1, 192.168.10.3 to Laptop 2, 192.168.10.4 to the printer, and 192.168.10.5 to the mobile phone.

All addresses are unique and within the valid host range.

Step 7: Verify the configuration.

I checked that every device has the correct subnet mask and a unique IP address. I also confirmed that the network address and broadcast address have not been assigned to any device.

Conclusion: The 192.168.10.0/24 network has been successfully divided into four equal /26 subnets. Each subnet has 64 total IP addresses and 62 usable host addresses, allowing devices to be organized into separate network segments.

NETWORK